

MODEL QUESTION PAPER

REVISION-2010
SUB CODE:3089

Reg. No.....
Signature.....

THIRD SEMESTER DIPLOMA IN POLYMER TECHNOLOGY
FUNDAMENTALS OF ELECTRICAL & ELECTRONICS

Time: 3 Hrs

Max.marks:100

PART –A

(Answer all questions in one or two sentences. Each question carries 2 marks)

- I. 1. Define rms value of an alternating current
2. What are the losses in a transformer?
3. Name the different methods of electric heating
4. Draw the symbols of NPN and PNP transistor
5. What is a transducer?

5 x 2 marks =10

PART –B

(Answer any five questions. Each question carries 6 marks)

- II 1. State the necessity of starter for starting a dc motor
2. Derive emf equation of a transformer
3. With a neat sketch explain the working of moving iron type instrument
4. With a neat diagram explain the pipe earthing
5. With a neat sketch explain the common emitter configuration of NPN transistor
6. Explain the working of LVDT
7. Explain the working of 3 phase induction motor

5 x 6

marks =30

PART –C

(Answer one full question from each module. Each question carries 15 marks)

Module - I

- III. 1. With a neat sketch explain the constructional details of a dc generator 10
2. Explain the methods of speed control of dc shunt motor 5

OR

- IV. 1. Explain with neat diagram the classifications of dc generator 10
2. List out advantages and disadvantages of electric drives

Module - II

5

- V. 1. A 10 kva, 1000/100 v, single phase transformer has a full load copper loss of 800 watts and a core loss of 400 watts. Compute the efficiency of the transformer at 0.8 p.f
(a) full load (b) half load

8

2. With a neat sketch explain the working of megger

7

OR

- VI.1. With a neat sketch explain the working of permanent magnet moving coil
8

Instrument

2. Explain the working of a transformer and its classification based on core
Construction

Module -III	7
VII.1.Explain eddy current heating	8
2. With a neat sketch explain plate earthing	7
OR	
VIII.1.Explain any one method of electroplating	8
2. Explain dielectric heating	
Module -IV	7
IX. 1.What is a capacitive transducer? With a neat sketch explain any one type of	
8	
capacitive transducer	
2. Explain the working of class A amplifier	7
OR	
X. 1.What is an inductive transducer? Explain any one type of inductive transducer	
8	
2. Explain the common base configuration of PNP transistor	7